OBJCTIVE:

The primary objective of the project, "A Digital Mental Health and Psychological Support System for Students," is to create an accessible, confidential, and user-friendly platform that helps students maintain and improve their mental well-being. Students frequently face academic pressure, examination stress, career uncertainty, social isolation, family expectations, anxiety, and other emotional challenges. Many students are unwilling or unable to seek professional help immediately due to fear of judgment, lack of awareness, limited counselling facilities, or concerns about privacy. This system is designed to bridge that gap by providing early-stage psychological support through a digital platform.The system aims to help students understand and monitor their emotional condition using features such as mood tracking, self-assessment questionnaires, digital journals, stress-management activities, and mental health awareness resources. By regularly recording their feelings and stress levels, students can identify emotional patterns and take positive steps toward self-care. The platform can also provide personalised suggestions such as relaxation exercises, meditation techniques, motivational content, and relevant self-help materials.Another important objective is to establish an easy connection between students and professional counsellors. Students can request counselling sessions or receive guidance without feeling exposed or uncomfortable. In critical situations, the system can display emergency support information and encourage users to contact qualified professionals or emergency services.The project also aims to promote awareness and reduce the stigma associated with mental health issues among students. It is not intended to replace doctors, psychologists, or emergency medical services; rather, it serves as a supportive tool that encourages early help-seeking, self-awareness, and access to appropriate professional support.

ABOUT THE PROJECT:

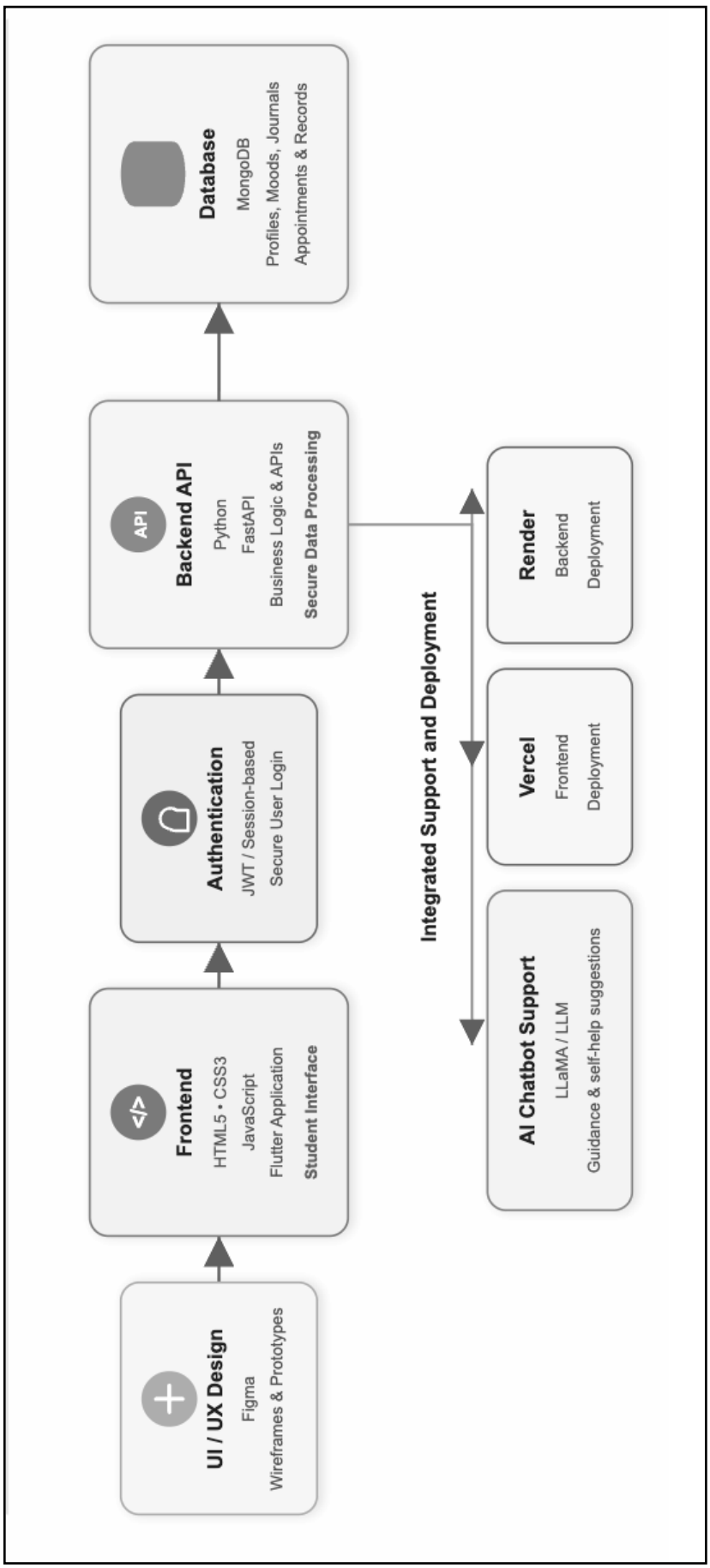
A Digital Mental Health and Psychological Support System for Students is a web-based or mobile-based platform developed to support the emotional and psychological well-being of students. In today's competitive academic environment, students may experience stress, anxiety, depression, loneliness, low confidence, and pressure related to examinations, career decisions, family expectations, and social relationships. Despite these challenges, many students do not seek help because of stigma, lack of awareness, limited counselling services, or fear that their personal information may not remain private.This project offers a confidential and convenient digital space where students can access mental health support at any time. The system allows users to create an account, maintain a personal profile, track their daily mood, write private journal entries, and complete basic well-being assessments. Based on the student's responses, the platform can suggest suitable self-help resources such as articles, videos, breathing exercises, meditation activities, and stress-management techniques.The proposed system also includes an AI-based chatbot that provides basic emotional support, answers general mental health-related questions, and guides users towards useful resources. Students can request or schedule counselling sessions with registered counsellors through the platform. Administrators can manage student accounts, counsellor details, appointments, and mental health resources.The system is designed to encourage students to recognize their emotional concerns at an early stage and seek appropriate support. It does not replace professional diagnosis or medical treatment; instead, it acts as a supportive bridge between students, self-help resources, and professional counselling services.

PROJECT FLOW:

1. Student registers and logs into the system.
2. The student completes an initial mental well-being assessment.
3. Based on responses, the system provides suitable self-help resources and recommendations.
4. Students can record daily moods, thoughts, and stress levels.
5. The chatbot offers basic emotional support and directs users to relevant resources.
6. Students can book appointments with available counsellors.
7. Counsellors can review student requests and provide guidance through the platform.
8. An administrator manages users, counsellors, resources, and reports.

METHODOLOGY:

- Requirement analysis
- System design using UML diagrams and database design
- Agile software development
- Module-wise implementation
- Unit testing and integration testing
- User acceptance testing
- Privacy and security validation


UI / UX Design
Figma
Wireframes & Prototypes
Frontend
HTML5 • CSS3
JavaScript
Flutter Application
Student Interface
Authentication
JWT / Session-based
Secure User Login
Backend API
Python
FastAPI
Business Logic & APIs
Secure Data Processing
Database
MongoDB
Profiles, Moods, Journals
Appointments & Records
Integrated Support and Deployment
Render
Backend
Deployment
Vercel
Frontend
Deployment
AI Chatbot Support
LLaMA / LLM
Guidance & self-help suggestions

MODULES:

1. **User Registration and Login Module**

   Allows students, counsellors, and administrators to securely create accounts and log in.

2. **Student Profile Module**

   Stores student details, preferences, emergency contacts, and mental health support preferences.

3. **Mental Well-Being Assessment Module**

   Provides questionnaires to understand stress, anxiety, mood, and emotional well-being levels.

4. **Mood Tracker and Digital Journal Module**

   Enables students to record daily moods, emotions, stress levels, and personal notes.

5. **Self-Help Resource Module**

   Includes articles, videos, breathing exercises, meditation guides, and stress-management tips.

6. **AI Chatbot Support Module**

   Offers instant general guidance, motivational messages, and links to relevant mental health resources.

7. **Counselling Appointment Module**

   Allows students to request and schedule sessions with counsellors.

8. **Admin and Reporting Module**

   Helps the administrator manage users, counsellors, resources, appointments, and anonymized usage reports.

TECHNOLOGIES USED:

- **Frontend – HTML5, CSS3, JavaScript:**

  Used to create the user interface of the system. HTML5 structures web pages, CSS3 provides styling and responsive design, and JavaScript adds interactive features such as forms, mood tracking, validation, and dynamic content.

- **Backend – Python and FastAPI:**

  Python is used for developing the server-side logic of the application. FastAPI is a modern Python framework used to build fast, secure, and scalable APIs for communication between the frontend, database, chatbot, and user modules.

- **Database – MongoDB:**

  MongoDB is a NoSQL database used to store user profiles, mood records, journal entries, counselling appointments, chatbot conversations, and resource information. Its flexible document-based structure is suitable for handling different types of student data.

- **Authentication – JWT / Session-Based Authentication:**

  Authentication ensures that only authorized users can access the system. JWT tokens or session-based login mechanisms are used to securely manage student, counsellor, and administrator accounts.

- **AI Chatbot – LLaMA or Other LLM:**

  A Large Language Model such as LLaMA can be integrated to provide basic conversational support, answer general mental health questions, suggest self-help resources, and guide students toward professional support when necessary.

- **Deployment – Vercel or Render:**

  Vercel can be used to deploy the frontend application, while Render can be used to host the FastAPI backend and related services. These platforms make the system accessible online.

- **Design Tools – Figma and Flutter:**

  Figma is used for designing user-interface layouts, wireframes, and prototypes before development. Flutter can be used if the project is extended into a cross-platform mobile application for Android and iOS.

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EXPECTED OUTCOME:

The project will deliver a secure, user-friendly platform that helps students monitor mood, complete well-being assessments, maintain journals, and access self-help resources. An AI chatbot will provide basic guidance, while the counselling appointment feature connects students with professional support. The system is expected to improve mental-health awareness, encourage early help-seeking, reduce stigma, and provide confidential support through secure authentication and data management.

CERTIFICATE

This is to certify that the project entitled **A Digital Mental Health and Psychological Support System for Students** is the result of my own work and has not been copied from any other source, including the Internet. Any resources, texts, or subject matter used during the development of this project have been duly acknowledged in this report.

This project report is submitted in partial fulfillment of the requirements for the second-year third-semester course of B.Tech.(CSE) with the code BCC 351, in accordance with the curriculum of Dr. APJ Abdul Kalam Technical University, Lucknow. I confirm that the content presented in this report is not being used for any other purpose and has not been submitted elsewhere.

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